

ASSESSMENT TOOL 3 – VISUALIZATION DESIGN ACTIVITY

Course Code: DSA0502 | Course Title: Query Processing for Data Science

CO Assessed: CO5 (Create meaningful visualizations using Pandas and Matplotlib)

Activities Presented: Activity 2 & Activity 8 Only

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Project Mode:	Individual Submission

Date of Submission:

ACTIVITY 2 – PLOT COMPONENT CHALLENGE

1. Aim

To identify missing plot components in an incomplete Matplotlib script and construct a properly annotated, clear, and readable line plot complete with a title, axis labels, legend, markers, and grid.

2. Incomplete Python Program

Below is the initial, incomplete Matplotlib program provided for the challenge. It lacks structural annotations, making the graph difficult to interpret.

```
import matplotlib.pyplot as plt

months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun']
sales = [120, 150, 220, 270, 310, 350]

plt.plot(months, sales)
plt.show()
```

3. Missing Components Identified

- **plt.title()** – Provides an overall descriptive title for the chart context.
- **plt.xlabel()** – Labels the horizontal axis to explain the independent variable (Months).
- **plt.ylabel()** – Labels the vertical axis to explain the dependent variable (Sales).
- **label=** and **plt.legend()** – Identifies the specific data series being plotted.
- **marker=** – Highlights individual data coordinate points clearly on the line.
- **plt.grid()** – Enhances visual tracking and readability across axes.

4. Corrected Python Program

The complete, executable Python program incorporating all essential visualization components is presented below:

```
import matplotlib.pyplot as plt

# Dataset definition
months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun']
sales = [120, 150, 220, 270, 310, 350]

# Create line plot with markers and color styling
plt.figure(figsize=(6, 3.5))
plt.plot(months, sales, marker='o', color='teal', linewidth=2, label='Monthly Sales ($)')
plt.grid()
```

```
# Add essential components
plt.title('Monthly Sales Performance (Activity 2)', fontsize=11,
fontweight='bold')
plt.xlabel('Months', fontsize=10)
plt.ylabel('Sales ($)', fontsize=10)
plt.legend(loc='upper left')
plt.grid(True, linestyle='--', alpha=0.6)

# Optimize layout and display
plt.tight_layout()
plt.show()
```

5. Expected Output

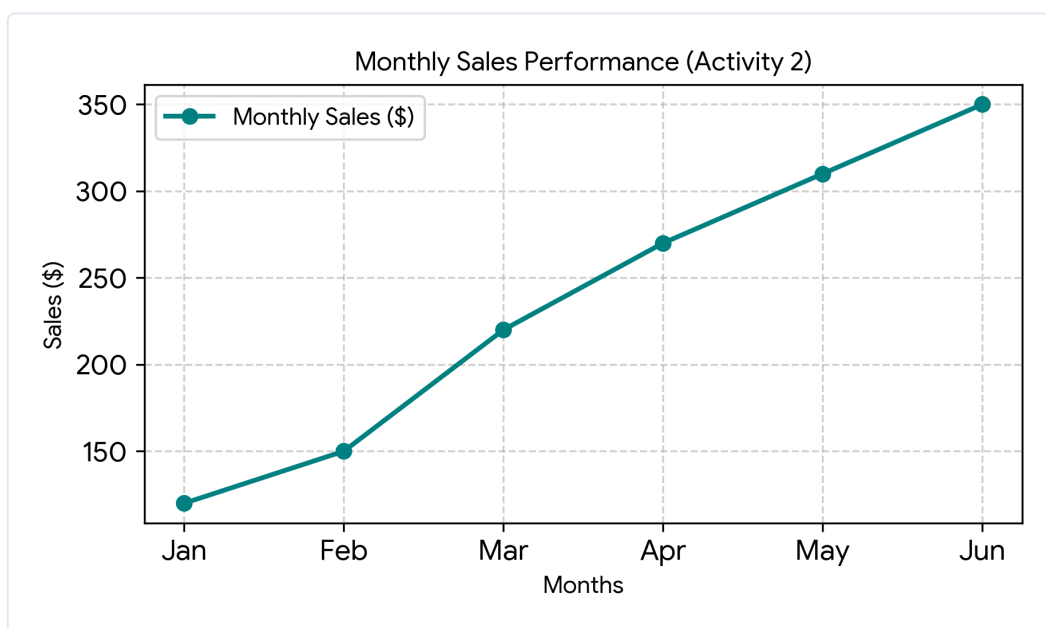


Figure 1: Corrected Matplotlib Line Plot with Essential Plot Components

6. Explanation

Adding a descriptive title and explicit axis labels immediately informs the viewer about the metrics being measured. Incorporating circular markers pinpoints exact monthly data intersections, while the dashed grid lines allow effortless tracking of values from data points to the axes. Finally, the legend clarifies the dataset being represented.

7. Result

The incomplete Matplotlib program was successfully corrected by adding all required visualization components, resulting in a professional and clear line plot.

ACTIVITY 8 – FREQUENCY VISUALIZATION

1. Aim

To create a categorical dataset representing department-wise employee distribution and visualize its frequencies effectively using a bar chart.

2. Dataset

A categorical dataset containing five distinct organizational departments and their respective employee counts is structured below:

Department	Employee Count
IT	45
HR	18
Finance	25
Marketing	30
Operations	22

3. Python Program

The complete executable Python script utilizing Pandas and Matplotlib to generate the frequency bar chart:

```
import matplotlib.pyplot as plt
import pandas as pd

# Define categorical dataset
data = {
    'Department': ['IT', 'HR', 'Finance', 'Marketing', 'Operations'],
    'Employee Count': [45, 18, 25, 30, 22]
}
df = pd.DataFrame(data)

# Create frequency bar chart
plt.figure(figsize=(6, 3.5))
plt.bar(df['Department'], df['Employee Count'], color=['#3498db', '#e74c3c',
'#2ecc71', '#f39c12', '#9b59b6'])

# Add labels, title, and grid
plt.title('Department-wise Employee Frequency (Activity 8)', fontsize=11,
fontweight='bold')
```

```
plt.xlabel('Department', fontsize=10)
plt.ylabel('Employee Count', fontsize=10)
plt.grid(axis='y', linestyle='--', alpha=0.6)

# Display plot
plt.tight_layout()
plt.show()
```

4. Frequency Chart Output

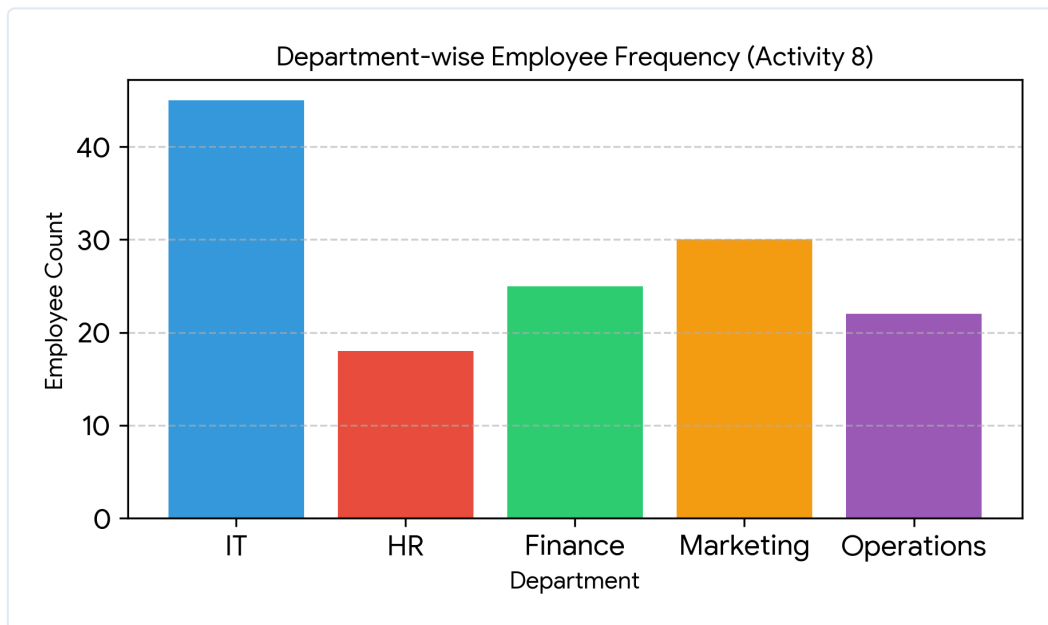


Figure 2: Department-wise Employee Frequency Bar Chart

5. Observations

- The **IT Department** has the highest employee frequency with **45 employees**.
- The **HR Department** has the lowest employee frequency with **18 employees**.
- The Marketing department maintains a moderate size with 30 employees, outnumbering Operations (22) and Finance (25).
- The bar chart format enables instant visual comparison of workforce distribution across distinct organizational categories without requiring manual data inspection.

6. Interpretation

A bar chart is ideally suited for categorical frequency data because discrete categories cannot be plotted on a continuous numerical scale. The length and height of each bar accurately encode the frequency or count of items within each respective category.

7. Result

Department-wise employee frequencies were successfully created and represented using a professional bar chart.